

Name: _____

Date: _____

SURDS, (ADDING AND SUBTRACTING SURDS)

LO: I can add and subtract surds by simplifying and collecting like terms.

Adding and subtracting surds

A **surd** is a root that cannot be simplified to a whole number (e.g. $\sqrt{2}$, $\sqrt{5}$). We can only add/subtract **like surds** (same number under the root).

Rule: $a\sqrt{n} + b\sqrt{n} = (a + b)\sqrt{n}$. Simplify surds first to find like terms.

Quick examples

●	$3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$	like surds
●	$8 + \sqrt{2} = 2\sqrt{2} + \sqrt{2} = 3\sqrt{2}$	simplify first
●	$7\sqrt{3} - 2\sqrt{3} = 5\sqrt{3}$	subtract like surds
●	$2 + \sqrt{3} = \sqrt{3}$	cannot add unlike surds
●	$3\sqrt{2} + 4\sqrt{3} = 7\sqrt{5}$	unlike surds stay separate

Key idea

Simplify each surd, then add or subtract the coefficients of like surds.

$$a\sqrt{n} + b\sqrt{n} = (a + b)\sqrt{n}$$

Only like surds can be combined

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – like surds

Simplify $4\sqrt{5} + 7\sqrt{5}$.

Same surd ($\sqrt{5}$)

Add coefficients: $4 + 7 = 11$
 $= 11\sqrt{5}$

$11\sqrt{5}$

Add the numbers in front of the same surd.

Example 2 – simplify then add

Simplify $12 + 27$.

$12 = (4 \times 3) = 2\sqrt{3}$

$27 = (9 \times 3) = 3\sqrt{3}$

$2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$

$5\sqrt{3}$

Simplify each surd first to reveal like terms.

Example 3 – subtraction

Simplify $50 - 18$.

$50 = (25 \times 2) = 5\sqrt{2}$

$18 = (9 \times 2) = 3\sqrt{2}$

$5\sqrt{2} - 3\sqrt{2} = 2\sqrt{2}$

$2\sqrt{2}$

Simplify, then subtract coefficients of like surds.

Example 4 – mixed expression

Simplify $2\sqrt{3} + 48 - 12$.

$48 = 4\sqrt{3}$; $12 = 2\sqrt{3}$

$2\sqrt{3} + 4\sqrt{3} - 2\sqrt{3}$

$= 4\sqrt{3}$

$4\sqrt{3}$

Simplify all surds, then collect like terms.

YOUR TURN – PRACTICE (deliberate practice)

1. Adding like surds

Simplify each expression.

- a) $2\sqrt{3} + 5\sqrt{3}$
- b) $6\sqrt{7} + 3\sqrt{7}$
- c) $5 + 4\sqrt{5}$
- d) $8\sqrt{2} + \sqrt{2}$

2. Subtracting like surds

Simplify each expression.

- a) $9\sqrt{2} - 4\sqrt{2}$
- b) $7\sqrt{3} - 3\sqrt{3}$
- c) $10\sqrt{5} - 6\sqrt{5}$
- d) $5\sqrt{11} - 5\sqrt{11}$

3. Simplify then combine

Simplify the surds first, then add or subtract.

- a) $8 + \sqrt{2}$
- b) $12 + \sqrt{75}$
- c) $50 - \sqrt{32}$
- d) $45 + \sqrt{20}$

4. Mixed expressions

Simplify fully.

- a) $3\sqrt{2} + \sqrt{8}$
- b) $2\sqrt{5} + \sqrt{45} - \sqrt{20}$
- c) $27 + \sqrt{48} - \sqrt{3}$
- d) $4\sqrt{7} - \sqrt{28} + \sqrt{63}$

5. Challenge simplification

Simplify each expression completely.

- a) $72 + \sqrt{50} - \sqrt{8}$
- b) $2\sqrt{12} - \sqrt{27} + \sqrt{75}$
- c) $98 - \sqrt{32} + \sqrt{18}$
- d) $3\sqrt{20} + 2\sqrt{45} - \sqrt{80}$
- e) $200 - \sqrt{50} + \sqrt{8}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$ ☐

Correct answer: _____

Reason: _____

b) $3 + \sqrt{5} = 8$ ☐

Correct answer: _____

Reason: _____

c) $12 = 2\sqrt{3}$ ☐

Correct answer: _____

Reason: _____

d) $8 + \sqrt{8} = 16 = 4$ ☐

Correct answer: _____

Reason: _____

e) You must simplify surds before adding them ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Simplify $75 + \sqrt{48} - \sqrt{108} + \sqrt{27}$. Express your answer in the form $a\sqrt{3}$.

Expression: _____

Simplified: _____

b) The sides of a triangle are 50 cm , 32 cm , and 8 cm . Find the perimeter in simplified surd form.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

